DUPLICATE RECORD CHECK

Sample Superstore Dataset

TASK 13 | DATA ANALYTICS INTERNSHIP

Task	Task 13 — Duplicate Record Check
Organization	Veda Technologies
Dataset	Sample Superstore
Tools	Python, Pandas, Jupyter Notebook / Google Colab
Deliverables	Duplicate report, cleaned dataset, audit note

This report documents the workflow for identifying exact duplicate records in the Sample Superstore dataset, reviewing potential duplicates, producing a cleaned copy, and recording the decisions in an audit note.

Results note: Actual duplicate counts have not been inserted because the dataset was not provided with this request. Run the notebook on your own dataset and fill in the results table before submitting.

1. Objective

The objective of Task 13 is to understand duplicate detection and data quality. The workflow checks for rows that match across all columns, reviews relevant combinations of business columns where appropriate, removes confirmed exact duplicates, and documents the outcome.

2. Dataset Overview

The Sample Superstore dataset commonly contains sales order, customer, product, sales, quantity, discount, and profit fields. The exact schema varies by version, so the notebook inspects the columns of the uploaded file before running checks.

A single Order ID may occur on several rows because one order can contain multiple product line items. Therefore, a repeated Order ID alone does not prove that a record is a duplicate.

3. Tools Used

Python is used to execute the workflow, Pandas handles tabular data and duplicate detection, and Jupyter Notebook or Google Colab provides an interactive environment for running and reviewing the analysis.

4. Methodology

Step	Action	Purpose
1	Load the source CSV or Excel file.	Read the dataset.
2	Inspect columns, types, and missing values.	Understand the data structure.
3	Flag exact duplicates across all columns.	Identify fully repeated rows.
4	Review relevant business-key combinations.	Find candidates for review without assuming they are duplicates.
5	Remove confirmed exact duplicate rows.	Create a cleaned copy.
6	Export report, cleaned data, and audit note.	Keep evidence of the process.

5. Duplicate Detection Logic

Exact duplicate rows can be flagged with the following Pandas expression:

```
df.duplicated(keep=False)
```

The `keep=False` option flags all rows belonging to a duplicate group. After reviewing the matches, `df.drop_duplicates(keep='first')` keeps the first occurrence and removes later exact copies.

Business-key checks must reflect the actual structure of the data. For example, Order ID by itself is usually not unique in a line-item sales dataset, so candidate matches should be inspected before any records are removed.

6. Results Summary

Metric	Value after running notebook
Original row count	_____
Exact duplicate rows flagged	_____
Exact duplicate rows removed	_____

Rows in cleaned dataset	_____
Exact duplicates remaining	_____
Business-key combination reviewed	_____
Potential business-key duplicate count	_____

Populate the table from the actual notebook output. Do not treat potential business-key matches as deleted rows unless they have been verified.

7. Deliverables

File	Description
superstore_duplicate_report.csv	Exact duplicate rows flagged for review.
superstore_cleaned.csv	Dataset after removing confirmed exact duplicate rows.
duplicate_check_audit_note.csv	Source file, original and cleaned row counts, duplicate counts, and validation details

8. Why Duplicate Checks Matter

Duplicate transactions can inflate sales and quantity totals, distort profit calculations, and affect customer or product counts. At the same time, deleting legitimate repeated values can damage the dataset. A careful workflow uses exact-row comparison, reviews candidate keys, documents decisions, and validates the cleaned output.

9. Conclusion

Task 13 demonstrates a repeatable duplicate-checking process for the Sample Superstore dataset. The workflow separates exact duplicates from potential business-key matches and provides outputs that support review and traceability. Final numerical findings should be added after the notebook is run.